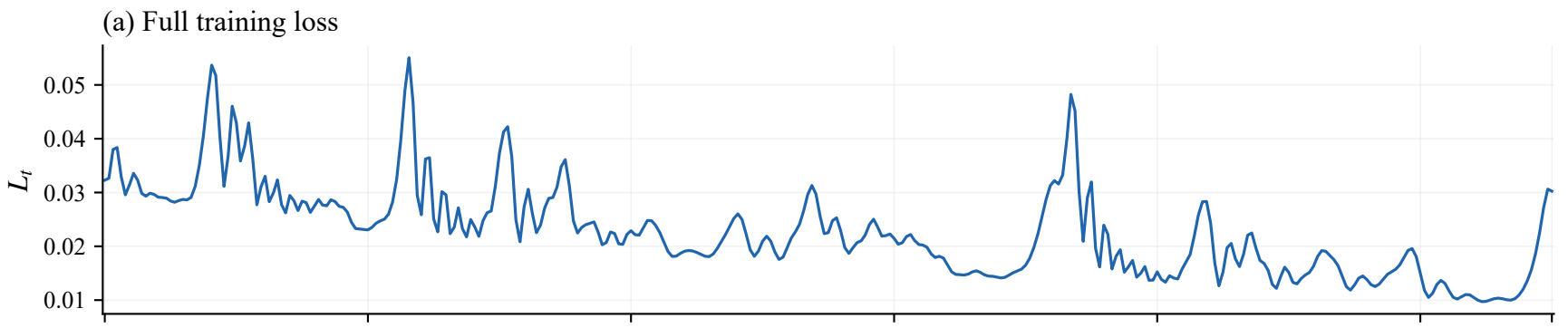
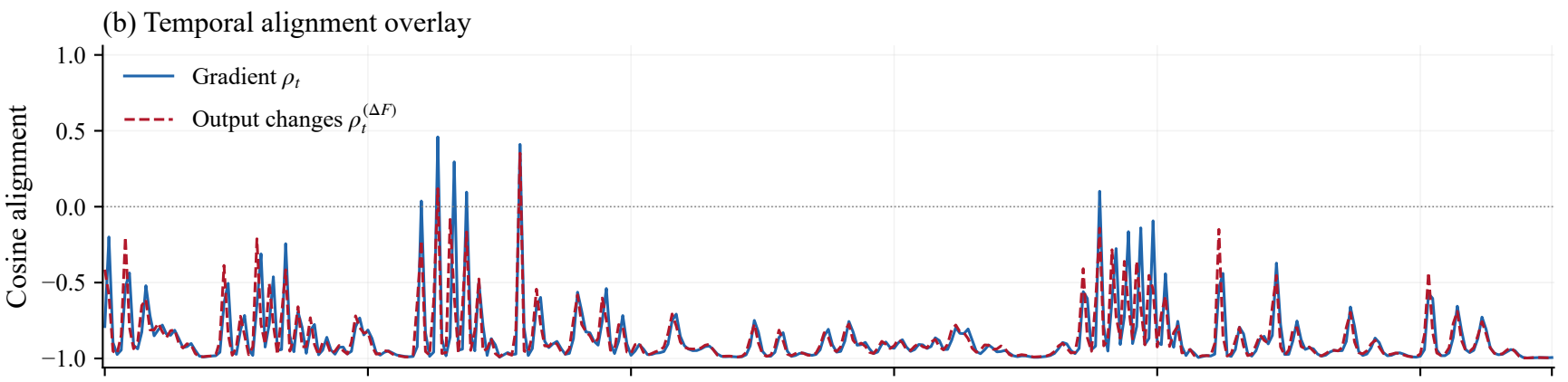
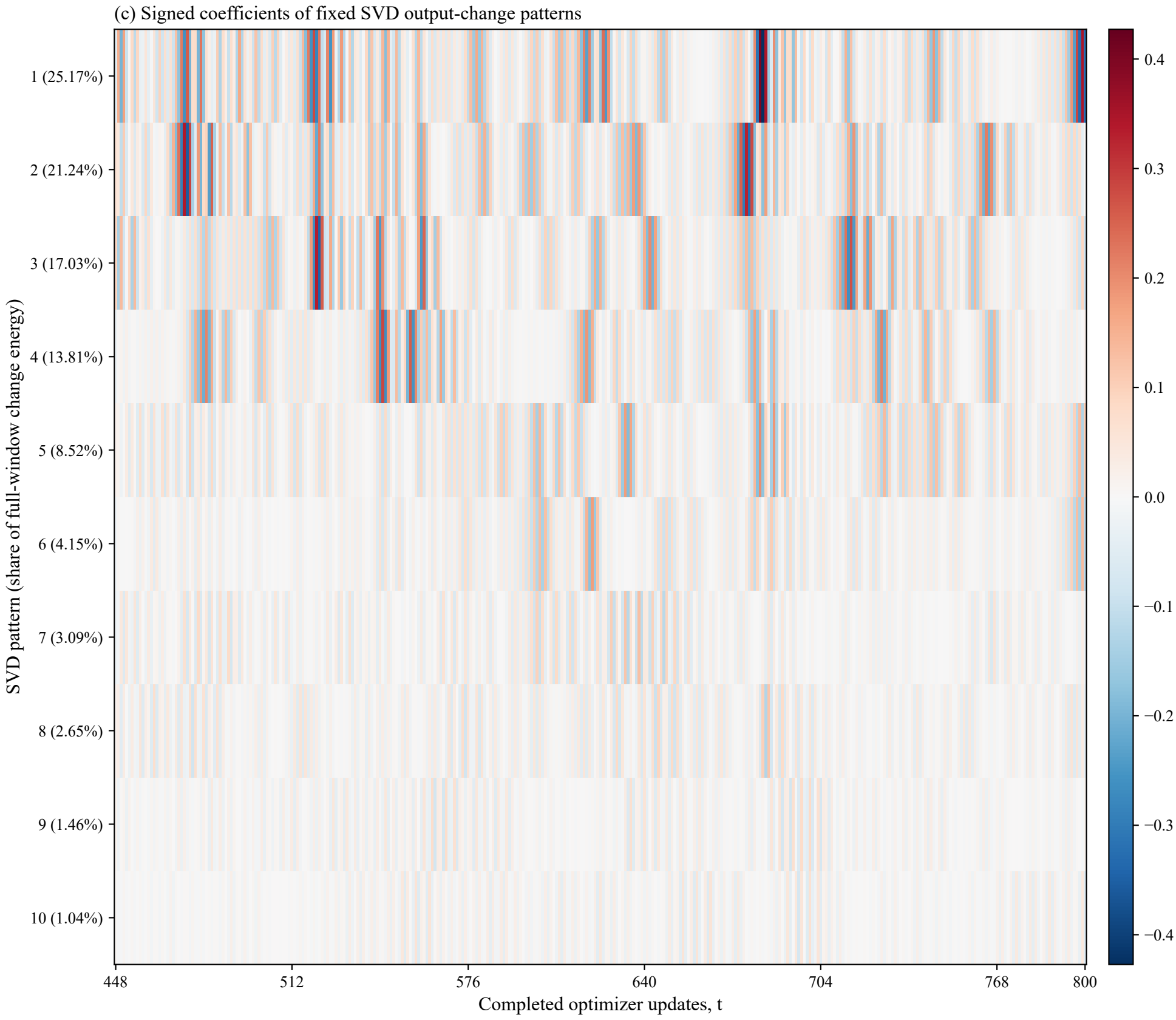
CIFAR-10 MLP: loss, temporal alignment and SVD output-change patterns

Unchanged Adam control | States 448-800 | Fixed SVD basis; all 5,000 examples retained; top 10 patterns capture 98.16% of change energy







$$\rho_t = \cos(g_t, g_{t-1}), \quad \rho_t^{(\Delta F)} = \cos(F_{t+1} - F_t, F_t - F_{t-1})$$

$$D_t = F_{t+1} - F_t \approx \sum_{k=1}^{10} a_{t,k} v_k; \quad v_k \in \mathbb{R}^{50000} \text{ is fixed across updates.}$$

No example averaging. Signs share one fixed orientation per pattern; colors use one symmetric scale.